



SMARTMOS: Modeling Subjective Audio Quality Evaluation for Real-Time Applications

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Abstract

Evaluating audio quality is a crucial task, with subjective listening tests being the gold standard. However, these tests are time-consuming intrusive and expensive, making them impractical for real-time applications like telecommunications. Despite efforts to develop automatic methods that match human listener fidelity, current approaches have limitations that hinder their use in real-world scenarios.

In this paper, we introduce SMARTMOS, a novel approach that addresses some of these gaps and enables accurate, real-time, non-intrusive, privacy aware audio quality assessment. We demonstrate the effectiveness of SMARTMOS through a case study on Noise Suppression (NS) and Packet Loss Concealment (PLC) modules. Furthermore, we show how semi-supervised learning techniques can be leveraged to build a joint MOS model that seamlessly covers both PLC and NS scenarios.

Index Terms: perceived audio quality, noise suppression, packet loss concealment, on-device models

1. Introduction

Measurement of quality is a very important step in developing any audio processing system, algorithms or modules. The metric to measure perceived audio quality can differ based on the focus of audio processing modules (APM). For a noise suppression system, one may be interested in measuring presence of residual noise and speech degradation due to aggressive NS, while for Packet Loss Concealment or Acoustic Echo Cancellation (AEC), one may want to assess the quality impact in the presence of artifacts and distortions to human speech. The question and design of the evaluation process are tailored to cater to these specific needs and hence, subjective tests usually yield the most accurate insights into the performance of the systems.

Objective evaluations such as PESQ [1], POLQA [2], and VISQOL [3] use a reference audio (typically referred to as intrusive technique) and aim to provide quality scores using hand-crafted features and metrics. These methods are easily scalable in certain scenarios, such as offline tests where clean references are available or can be synthetically generated. However, they are not suitable for scenarios without clean references, whether online, offline, or in product deployments. Subjective evaluations involve human labeling of processed audio, typically on a scale of 1-5 (1 being worst and 5 being best), to obtain opinion scores. This process is conducted either in a controlled environment or through crowd sourcing, following ITU-T recommendations [4, 5, 6]. To ensure reliability, ratings from multiple raters are obtained for each audio sample and statistically aggregated to obtain the mean opinion score (MOS). While subjective scores are considered the gold standard for audio quality assessment, obtaining them can be expensive and not scalable

for real-time scenarios.

Measuring audio quality in real-time applications, such as telecommunication, requires on-device predictions without the need for a reference signal. This is because these applications involve multiple APMs, and assessing audio quality at different points after each APM is crucial. Therefore, the algorithms used must be low-footprint (low on memory, latency and computation) and not interfere with the primary application. To address this challenge, ML-based techniques have been employed to model the subjective MOS without relying on a reference signal.

Self-supervised techniques [7, 8] aim to enhance the generality of model predictions using architectures like Wav2Vec [9] and HuBERT [10], and fine-tune the model for MOS predictions. However, these models have high computational and memory requirements, making on-device deployment challenging. SESQA [11] and MOSRA [12] provide overall audio quality using semi-supervised learning/Teacher-Student approaches for unlabelled data, but do not offer scores for artifacts related to specific APMs. Models such as [13, 14, 15, 16, 17] provide quality measurement for specific audio processing tasks, but using them require multiple models in the telecommunication pipeline. Models such as [18, 19, 20] predict audio quality from human ratings for various artifacts across audio processing type and provide ratings for multiple perceptual quality dimensions and overall quality. However, this requires obtaining human labels for every audio in the dataset along all the quality dimensions.

In this work, we present a single low-footprint on-device model that predicts human subjective ratings for NS and PLC APMs using semi-supervised training techniques. Unlike [18, 20], our model provides specific scores for each APM, with datasets collected separately and labeled specifically for each module. While we train models specifically per APM, we aim for generalisability within rating these APM modules. While the subjective nature of the task limits our ability to use other test sets and quality prediction models for direct comparisons, we provide real-world rating scenarios (Figure 4) that demonstrate the generalisability. We provide details on developing the joint model and compare its performance to standalone PLC and NS models. Additionally, we discuss techniques used to deploy this model on-device without CPU overloads using streaming inference and a state machine based sampling technique.

2. Dataset

A labeled dataset is one of the most important components for development of SMARTMOS. Our data collection/labeling process is designed with two major requirements: 1) To guarantee that the scores obtained across various sessions and raters are

comparable and normalized. This process control ensures that models trained using our dataset are capturing the quality scores and not just the variability/noise in the scoring process. 2) Maximize the diversity in the processing, scenarios and locales.

To ensure consistency and accuracy in their ratings, the raters were provided with a set of golden audio clips rated by experts that served as reference points for S, N, and O MOS targets. These golden clips were used to calibrate the raters at the start of each session, helping them to establish a common understanding of the desired quality standards. In addition to the calibration exercise, each evaluation set of 15 audio clips included up to 2 golden reference clips, which were randomly inserted among the other clips. However, these golden references were not explicitly identified as such to the raters, allowing us to assess their ability to consistently rate the clips without bias. If the raters failed to accurately score these golden references, the entire evaluation set was discarded to maintain the integrity of the rating process.

We built an in-house dataset to assess audio quality in various scenarios, including NS, AEC, and PLC. In this paper, we focus on a subset of the dataset that features speech audio clips with a single primary speaker. Other special scenarios, such as multiple speakers or equalization, is not the primary focus of this work. The overall collection comprises approximately 400k labelled audio clips, spanning eleven languages. For the NS related dataset we included audios with different noise types and levels processed by several in-house and third party NS algorithms. We also included unprocessed noisy data to get scores in different ranges. Similarly, for PLC dataset, we included audios with different packet loss rates processed by different PLC algorithms. Each utterance is typically 10 seconds long and is evaluated by multiple trained raters (ten per clip). During a session, raters are presented with only one task (e.g., either NS or PLC). To prevent listener fatigue, a limit is placed on the number of sessions that could be assigned to each rater.

For the dataset related to noise suppression APM, we aim to capture not only the overall audio quality but also the quality of the main speaker and the impact of noise. To achieve this, we design questions that specifically focus on each aspect, including Speech MOS (SMOS) for the quality of the main speaker, Noise MOS (NMOS) for the quality as impacted by the presence of noise, and Overall MOS (OMOS) for the overall audio quality. For the PLC APM, we label a smaller amount of data (1/3rd the size of NSMOS data) instructing raters to score these utterances solely for overall quality (i.e., OMOS).

For each processing type, we divide the dataset into training, validation, and testing sets in a 70:15:15 ratio, ensuring that the distribution is maintained across languages and locales. We also ensure that the speakers and utterances in each group are unique, and the test data is held back for final evaluations. The results presented in this paper are based solely on the test data.

3. Model Training

All the SMARTMOS models presented in this paper are trained to predict quality scores for an audio clip that is around 10 seconds. The model's inputs are log-mel features computed over a 20 ms frame that is generated with 10 ms overlap. We use 1024 size FFT and ablate over the number of mel-banks, details of this experiment is captured in Section 5. All the models use Adam optimizer and MSE loss for training. We use the mean of the ratings from all raters per audio clip as the label for training and testing the models.

3.1. Small vs Large Models

Standalone small models are trained to achieve low memory and computational efficiency for both Noise Suppression MOS (NS-MOS) and Packet Loss Concealment MOS (PLCMOS). The model architecture is similar to DNSMOS [14], with the number of outputs of the model varied according to the processing type. It is observed that replacing the global max pool layer with a global average pool layer yields improved results in this setup. These group of models use 40 filter banks. Due to the limited availability of PLCMOS data, transfer learning techniques are employed, wherein the model is initially trained on the overall dataset and subsequently fine-tuned with the PLC dataset

To further investigate the impact of model architecture on performance, larger SMARTMOS models are trained to examine the effects of increased receptive field and improved context utilization. A secondary objective is to determine whether these larger models could be leveraged to augment the training set with semi-supervised labels. This investigation entails the use of higher-resolution input, incorporating larger input filter banks (e.g., 40, 80, and 120), as well as deeper and wider network architectures. Furthermore, SMARTMOS models with Emformer layers are trained to compare and contrast the attention mechanism with convolutional neural networks (CNNs) for the tasks, providing insight into the relative strengths of each approach.

Emformer [21] is a transformer architecture variant that uses a fixed-length context window to attend to the input sequence, rather than attending to the entire sequence at once. In Emformers, the left and right context lengths determine the number of time steps the model considers before and after the current time step when processing the input sequence. The segment size represents the chunk of the input sequence processed in the current time step. These parameters primarily impact the analysis context for the Emformer, while the number of layers affects long-range dependencies. Feature stacking is a technique used to increase context per time step by concatenating past and future time step features with the present time step. The receptive field can be controlled using both feature stacking and context lengths.

3.2. Joint On-device Models

SMARTMOS models trained for on-device deployment are based on CNN architecture and our goal is to train a single model to predict both the PLC and NS MOS scores. As explained in section 2 PLC (OMOS only) and NS data set have different types of MOS labels which complicates the joint model training.

We try a semi-supervised approach to overcome this label mismatch issue by training a single SMARTMOS model with four output targets: three outputs for NS scores and one for predicting the OMOS for PLC. Unlike traditional semi-supervised learning methods [22, 23, 24], which use a pre-trained model to generate same type of labels for unlabelled data, we use one model to generate labels for the dataset of another model, effectively bridging the gap between datasets. As shown in Table 1, we use standalone Emformer models trained for NS and PLC cases as teachers to predict pseudo-labels for the other dataset, thereby creating a cross model knowledge transfer. This approach is non-trivial because the artifacts penalized by the MOS model during scoring differ between APMs. Scores do not correlate well when using NSMOS for PLC APM scoring and vice versa, with correlations as low as 50%.

Table 1: Labels used for each dataset to train four score joint model.

Data	OMOS NS	OMOS PLC	N and S MOS
NSMOS	Human ratings	PLCMOS Emformer	Human ratings
PLCMOS	NSMOS Emformer	Human ratings	NSMOS Emformer

4. On-Device model deployment

The SMARTMOS model is developed with the aim to deploy it in VoIP apps so we can understand, monitor and improve quality of audio for Meta Communication services. To ensure seamless calling experience for the user we have to ensure the models are running efficiently on-device and have minimal impact on memory, CPU and battery. These aims are achieved by implementing a smart strategy to select audio segments, rate limiting number of MOS evaluations during a call and restructuring the SMARTMOS model to prevent single shot processing of 10 seconds audio data.

4.1. Audio Segment Selection

The selection of the best available 10 second segment of audio is done by using a voice activity detector(VAD) in combination with a small state machine as shown in Figure 1. The sampler operates with predetermined action in one of the three states: "speech", "silence", and "null". While in "speech" state the sampler actively buffers the frames of processing if voice activity is present otherwise it transitions to "silence" state. The sampler only invalidates the buffered data if "silence" detected for X seconds and the sampler then transitions to "null" state. As seen from the figure we build stickiness in each state to prevent unnecessary invalidation of accumulated audio. We also use the sampler to rate limit the MOS evaluations to Y in Z minutes. We initialize the sampler from the "null" state at the start of the call. The values of parameters are X , Y and Z are tuned heuristically.

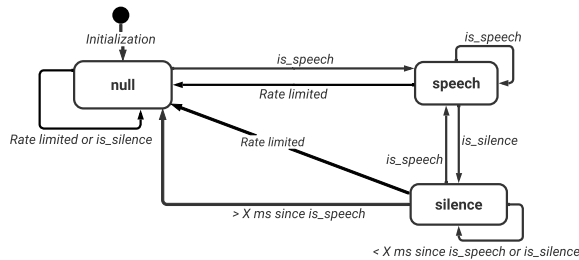


Figure 1: Smart Sampler Operation.

4.2. Reducing CPU Spikes

Evaluating over 10 second segment of audio in single shot causes unnecessary spike in CPU. To address this, we split the model into two parts: encoder, which contains all the convolution layers and predictor, which has the fully connected components as seen in the Figure 2. The input audio segment is split into chunks (e.g. 0.1 seconds each) and processed through the encoder at a slower rate, the embedding from the encoder are accumulated and processed in single shot when the embedding for full segments are available.

We compare the CPU usage of the streaming model with

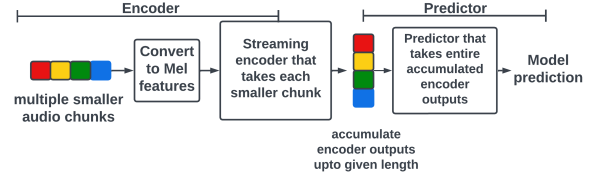


Figure 2: Streaming inference with encoder predictor structure

state machine based sampling against the non-streaming model on-device. As shown in Figure 3, the modified model's CPU usage remains within a range and predicts Y times every Z minute, whereas the non-streaming model peaks every 10 seconds, which is not desirable for a real-time system.

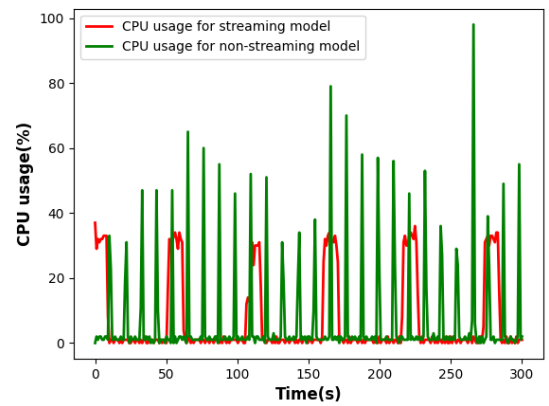


Figure 3: CPU usage for streaming model w/ sampler vs non-streaming model, where $(X, Y, Z) = (2s, 1, 1min)$

5. Experiments and results

In the following subsections, we analyze the performance of the models through a series of experiments. Specifically, we employ Pearson Correlation Coefficient (PCC) [25] to compare the correlation between the mean scores from subjective evaluations and the model's MOS predictions per audio clip.

The SMARTMOS models are trained to mimic our subjective evaluation process and predict the scores similar to the trained raters that we used during our study.

These scores might be different than the ones predicted by other SOTA models like DNSMOS or TorchSquim [26]. This mismatch does not mean one model is superior to another, it just highlights the inherent variability of subjective testing and each model is trying to predict/conform to its training dataset or achieve a particular objective.

For example, while the DNSMOS model has a PCC of 69.2% when evaluated on our test dataset, SMARTMOS achieves a PCC of 92% for OMOS. A direct comparison of the model's capabilities and presenting this number to readers will incorrectly reflect poorly on capabilities of the DNSMOS models.

We also make a deliberate choice to not present objective scores such as PESQ in this study because these metrics don't correlate well with human ratings [14].

Table 2: PCC % for best CNN and Emformer models on held out test sets.

MOS Type	Num Param	OMOS NS	OMOS PLC	NMOS	SMOS
PLC CNN	45K	NA	85.1	NA	NA
PLC CNN	68K	NA	84.1	NA	NA
PLC EMF	1M	NA	85.5	NA	NA
NS CNN	45K	90.8	NA	87.8	89.7
NS CNN	68K	91.2	NA	89.1	89.2
NS EMF	1M	92.9	NA	91.6	91.5
Joint CNN (3S)	45K	91.5	82.3	89.7	89.8
Joint CNN (4S)	45K	91.9	84.6	90.1	90.3

5.1. Correlation to Human Opinion Scores

Table 2 presents the correlations of the top-performing CNN and Emformer models. The results are calculated on the held-out test sets corresponding to each dataset. Notably, the joint MOS model with four scores outperforms the standalone NS-MOS model for the NS task and achieves comparable performance to the standalone PLCMOS models for the PLC task. This improvement is attributed to the inclusion of semi-supervised labels from the best offline models. In contrast, the joint three scores model shows a degradation in the OMOS PLC correlation. This regression might be caused by (1) data imbalance between the two datasets or (2) differences in the quality prediction task between the processing algorithms. The Emformer models do exhibit better performance which is accompanied with larger model size.

Table 3: PCC % for Emformer NSMOS models with just OMOS on heldout test sets. LMEL denotes number of log-mel filterbanks used and L, R denotes left and right context respectively

LMEL	Num Layers	[L,R]	OMOS
120 (FS)	5	[10, 1]	92.2
	2	[10, 1]	91.7
	2	[4, 1]	91.5
	3	[10, 1]	92.6
40 (FS)	2	[10, 1]	91.5
	10	[4, 1]	92.1
40	3	[4, 1]	90.1

Table 4: PCC % for best Emformer NSMOS models with all 3 scores on held out test sets. LMEL denotes number of log-mel filterbanks used.

LMEL	Num Layers	[L,R]	OMOS	SMOS	NMOS
40 (FS)	3	[10, 1]	92.9	91.5	91.5
120 (FS)	3	[10, 1]	92.9	91.5	91.6
	9	[10, 1]	92.7	91.2	91.7
	3	[30, 2]	92.8	91.4	91.4
	9	[30, 2]	92.2	91.1	90.7

5.2. Emformer Ablations

We conduct ablations on the Emformers by fixing the segment size to ten but varying the size of left and right context and the number of Emformer layers. Each input feature consists of four stacked frames. We perform two experiments for NSMOS

dataset, one with OMOS score predictions (Table 3) and another that predicts all three sets of MOS scores. (Table 4).

The gap between the performance of the CNN and Emformer for both NS and PLC cases is not significantly large (Table 2). We hypothesize that the local context is more important for predicting quality than a larger receptive field. Therefore Emformers do not benefit a lot over the CNN's even when they are able to attend to the entire feature dimension over a larger context. The Emformer benefit more from the stacked features as compared to the CNN models. However, increasing the receptive field through feature stacking or context lengths comes at a high computational cost. The best-performing Emformers have nearly one million parameters (Table 2), making CNNs a better choice for on-device deployments due to their lower computational requirements. Additionally, We also see that predicting the three scores together improves OMOS correlations.

5.3. Correlation in Real-Time Calls

We deployed SMARTMOS models on-device using techniques mentioned in section 4 to generate audio quality scores that focused on measuring the performance of noise suppression (NS) modules in two calling apps. The score generation was done in real-time to maintain user privacy, covering approximately 200k calls per app. The OMOS NS scores from SMARTMOS models were reported back along with user surveys. When users rate call quality, they often use star ratings that reflect a mix of issues caused by both network and audio processing algorithms.

These comparisons demonstrate the generalisability of the MOS model in rating the APM (in this case NS), as real-world calls involve diverse background noise scenarios not seen during training. They also have different on-device NS APMs (hardware/software) depending on the user's calling device. Our experiments found calls with higher quality scores tend to have fewer noise related complaints and vice versa. This suggests that our SMARTMOS models are effective at measuring call quality issues specific to an APM. Additionally, this is also significant because real-world surveys are often sparse, and the SMARTMOS models serve as a proxy to fill gaps in the surveys, providing valuable insights into real-world calls.

6. Conclusion and future work

We train SMARTMOS a single joint audio quality prediction model for rating both Noise Suppression and Packet Loss Concealment APMs in real-time telecommunication, using semi-supervised techniques. We demonstrate techniques that make the SMARTMOS model work on-device without increasing CPU load and show correlation with metrics from real-time calls. Additionally, we find Emformers with larger receptive fields slightly outperform CNNs but at a high computational cost, making CNNs a better choice for on-device deployments.

Future work includes expanding SMARTMOS models to rate other modules like AEC and experimenting with ensemble techniques to model the rating process as a whole, providing insights into how hard a particular audio is to rate.

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